Team Project (Team)

7.2/8 Points

Due: Sat Jul 19, 2025 9:00am

Attempt 1

Review Feedback
SUBMITTED: Jul 18, 2025 10:49pm

Attempt 1 Score:
7.2/8

View Feedback

Anonymous Grading: **no**

Unlimited Attempts Allowed

Details

Overview:

This project provides practical experience in statistical analysis, preparing you for real-world data scenarios. Your task includes exploring and preprocessing the data, applying appropriate tools, and preparing a report detailing your findings.

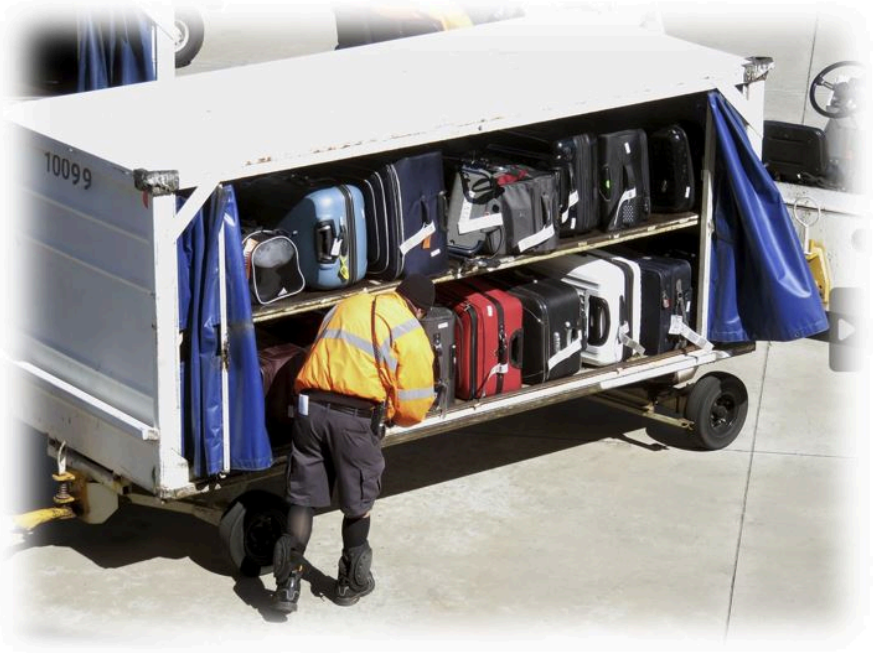
Instruction:

- The report must begin with an **executive summary**.
- The report is **four pages maximum**; you may include up to two additional pages of exhibits to support your responses. All questions listed in this document must be addressed.
- **You are allowed to use AI tools to support your analysis**. However, to maximize your learning experience, you should not rely on them to directly solve the assignment. Instead, use these tools for guidance, such as finding specific Excel formulas or programming code for certain tasks.
- Your submitted work **must be typed**. Acceptable file formats: **PDF**.

Grading/Rubric:

Your work will be evaluated based on the clarity and comprehensiveness of your executive summary, the depth and accuracy of your analysis, and overall adherence to the assignment instructions/guidelines.

Background:



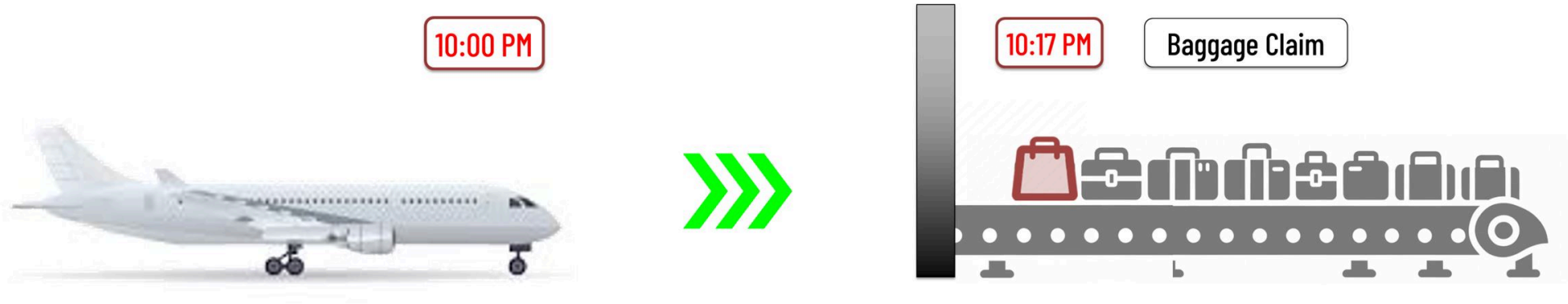
Baggage Unloading Process in Airline Operations

Upon the arrival of an aircraft at its destination, a coordinated sequence of operations is set in motion to ensure the safe and efficient disembarkation of passengers and their baggage. As the plane docks at the gate, ramp agents spring into action, positioning wheel chocks to secure the aircraft, connecting auxiliary power units for ground power and air conditioning, and installing a tail stand to prevent the aircraft from tipping.

Subsequently, a belt loader is aligned with the cargo hold, initiating the transfer of bagage. The luggage is sorted onto mobile carts, distinguishing between items destined for baggage claim and those in transit to connecting flights. Baggage meant for retrieval is transported to the airport's bag belt system, where it is manually loaded into a system ond are routed to the correct bag carousel, where they are scanned as they enter baggage claim.

In this analysis, you will focus on the "**baggage delivery time**" which is defined as the duration between a flight's arrival time (data column "**ActualArrival**") and the time that the last bag enters the baggage claim area (data column "**LastBagDropTime**"). Understanding the key

characteristics of baggage delivery time is essential for efficient scheduling, planning, customer satisfaction, and overall performance optimization.



Data Set:

[This data set (https://canvas.uw.edu/courses/1810306/files/136197016?wrap=1)](https://canvas.uw.edu/courses/1810306/files/136197016/download?download_frd=1) (https://canvas.uw.edu/courses/1810306/files/136197016/download?download_frd=1) contains information related to the baggage unloading process from actual flights. As is often the case with real-world data, this data set may include errors and inconsistencies. Hence, it is imperative to conduct a thorough examination of the data set prior to initiating any analytical procedures.

Questions:

Data Pre-processing

1. What issues and/or discrepancies do you observe in the data set? How pervasive are these issues?

2. How do you address these issues in the data set? What statistical tools or software is used for these tasks?

Data Analysis & Discussions

3. How would you describe the variable “baggage delivery time”? Utilize relevant statistical measures and/or visualization tools to effectively summarize and describe the data. If applicable, identify and report any anomalies or unexpected patterns in your data summary.

4. What is the probability that the baggage delivery time for a random flight is less than 21 minutes?

- Derive answers based on two approaches: Empirical: Use relative frequencies to directly answer the question. Theoretical: Select one probability distribution that best fits the data set (provide a brief justification for your choice) then use it to help answer the question.
- For both approaches, clearly support the final answer with relevant calculations. If applicable, discuss statistical tools or software used for these tasks.

5. Based on your analysis in the previous question, do you observe any meaningful difference when comparing the outcome derived from empirical methods vs. theoretical approaches? Discuss the advantages & disadvantages of the empirical and theoretical approaches.

Summary & Reflection

Describe the key insights and their implications, along with a discussion about the challenging aspects you've encountered when working with real-world data in this project.

	File Name	Size	
	BUS AN 51...eport.pdf	170 KB	Download (https://canvas.uw.edu/files/136872871/download?download_frd=1&verifier=E4lZJUAnAQr9c7aVi9wtSqxDWIk2K031rghBUe0D)